

MINI PROJECT – LED CHASER ASSEMBLY

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A report of an LED Chaser Kit Soldering & Circuit Analysis
EENG 1910.001

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I. ABSTRACT

This report describes the design, construction, and testing of an LED Chaser circuit built from a kit using a printed circuit board. The purpose of this project was to introduce basic circuit construction techniques, component identification, soldering methods, and troubleshooting practices. The circuit uses a 6 V power supply and a timer-based control system to sequentially illuminate ten LEDs in a repeating pattern.

The circuit consists of resistors, capacitors, integrated circuits, and light-emitting diodes arranged on a printed circuit board. A timer integrated circuit generates pulses which are sent to a counter integrated circuit that controls the order in which the LEDs turn on. The LEDs light in sequence, creating the visual effect of a moving light pattern.

During construction, several issues occurred including incorrect component polarity, poor soldering connections, and unintended electrical paths caused by solder bridges. These problems required multiple testing attempts and corrections before the circuit functioned properly. The board required three full trials before successful operation was achieved.

This report explains the construction process, theory of operation, mathematical principles, and troubleshooting steps involved in completing the LED chaser circuit.

II. INTRODUCTION

The LED Chaser mini project was designed to introduce students to basic electrical engineering concepts through hands-on circuit construction. The project required assembling a printed circuit board using through-hole components and soldering techniques. The completed circuit produces a repeating sequence of illuminated LEDs using a timer and counter system.

The project emphasized several important engineering skills including:

- Reading circuit diagrams
- Identifying electronic components
- Understanding polarity
- Soldering components
- Testing circuits
- Troubleshooting failures
- Understanding signal flow

The LED chaser kit utilizes a timer integrated circuit and a counter integrated circuit to create a sequence of pulses that control the illumination of ten LEDs. The result is a repeating pattern that appears as a moving light.

This project demonstrated how individual electronic components work together to form a complete and functional electrical system.

III. PROJECT OBJECTIVES

The main objectives of this project were:

- Construct a working LED chaser circuit
- Learn proper soldering techniques
- Understand component polarity
- Identify common circuit components
- Interpret circuit diagrams
- Test and troubleshoot a circuit
- Understand timer-based circuits

- Observe sequential logic operation

IV. COMPONENTS AND MATERIALS

The LED chaser circuit consisted of the following components:

Power Source:

- 6 V DC Voltage Supply

Integrated Circuits:

- 1x 8-pin Timer IC (555 Timer)
- 1x 16-pin Counter IC

Resistors:

- 11x 1 k Ω resistors
- 1x 10 k Ω resistor
- 1x UR 500 k Ω variable resistor

Capacitors:

- 2x 10 μ F electrolytic capacitors

Output Components:

- 10x Light Emitting Diodes (LEDs)

Other Materials:

- Printed Circuit Board (PCB)
- Solder
- Battery Leads

Each component serves a specific function within the circuit and must be placed correctly in order for the circuit to operate.

V. THEORY OF OPERATIONS

The LED chaser circuit operates using a timing signal that controls a counting circuit. The timer produces a repeating electrical pulse that advances the counter. Each output of the counter corresponds to one LED.

The timer integrated circuit operates in astable mode, meaning it continuously switches between high and low voltage states. This produces a square wave signal.

The counter integrated circuit receives the pulse signal and activates its outputs sequentially. Each output powers a different LED, creating a chasing effect.

The capacitors control the timing of the pulses, while the resistors control current flow and brightness.

The LEDs emit light when current flows through them in the correct direction. If polarity is reversed, the LED will not function.

VI. CIRCUIT OPERATION

The circuit begins operating when power is supplied by the 6 V battery pack connected to the VCC and GND terminals. Once power is applied, the NE555 timer starts generating pulses. These pulses are created using the capacitor, resistors, and the 500k potentiometer, which controls how fast the pulses are generated. Adjusting the potentiometer changes the speed of the LED sequence by changing how quickly the capacitor charges and discharges.

The pulses from the timer are sent to the CD4017 counter integrated circuit, which controls which LED turns on. Each time the counter receives a pulse, it moves to the next output pin. Each output pin is connected to an LED through a resistor, which limits the current so the LEDs do not burn out.

This process is shown in the IPO (Input–Process–Output) diagram. The input is the electrical power from the battery and the adjustable setting from the potentiometer. The process is carried out by the NE555 timer and CD4017 counter, where pulses are generated and then counted in sequence. The output is the LEDs turning on one at a time, creating the chasing light pattern.

The network diagram shows how all the components are physically connected on the circuit. The flowchart shows the step-by-step operation, starting from power being applied, then generating pulses, advancing the counter, and lighting each LED in order. The state diagram represents each LED as a different state, with the circuit moving to the next state every time a pulse is generated.

As long as power is supplied, the counter keeps cycling through the outputs, and the LED pattern repeats continuously.

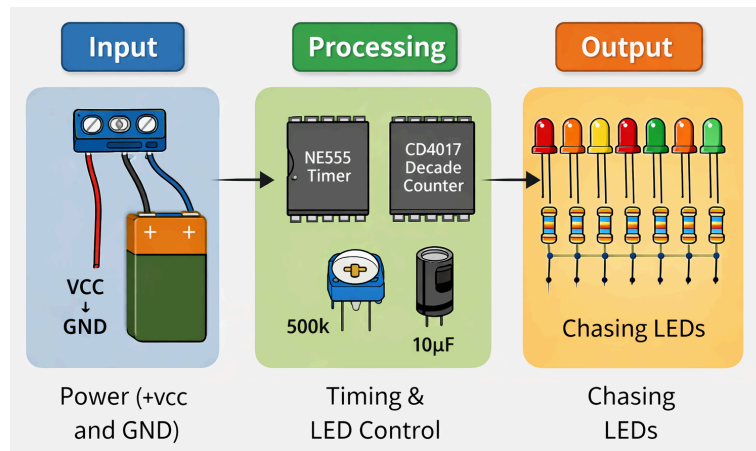


Figure 1. IPO Diagram – LED Chaser

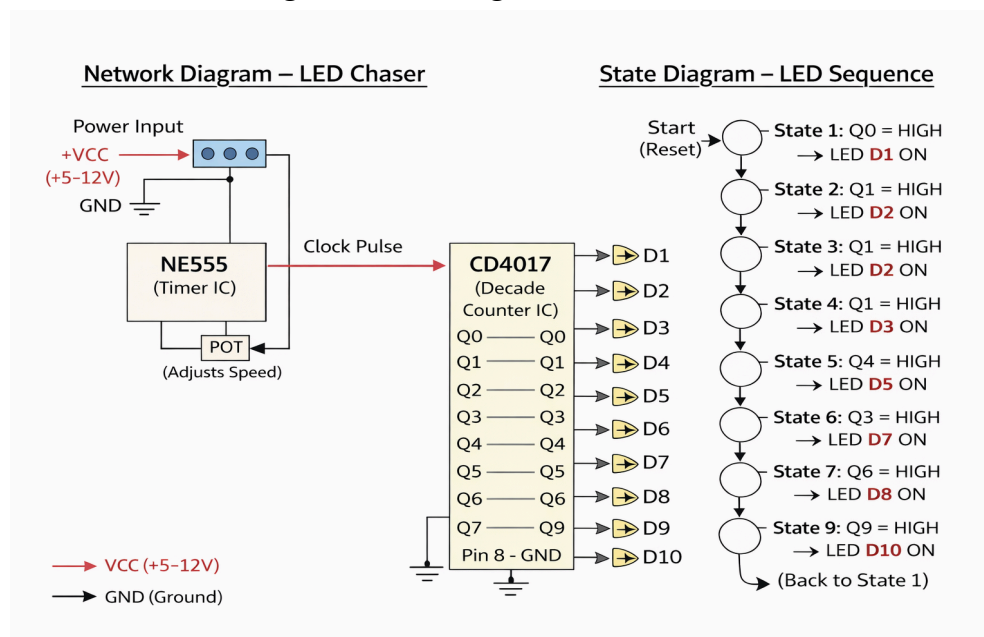


Figure 2. Network & State Diagram – LED Chaser

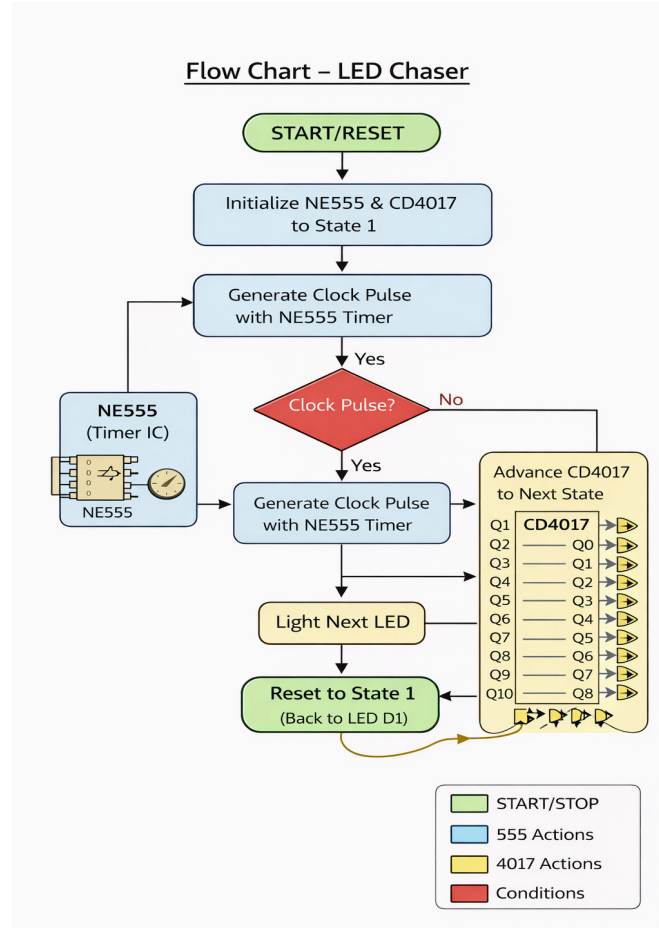


Figure 3. Flow Chart – LED Chaser Operation

VII. MATHEMATICAL ANALYSIS

The circuit operates using a 6 V power supply and resistors to control current through the LEDs.

Ohm's Law describes the relationship between voltage, current, and resistance:

$$V = IR$$

Each LED uses a 1 kΩ resistor to limit current. Assuming a typical LED voltage drop of 2 V:

$$I = (V_{\text{source}} - V_{\text{LED}}) / R$$

$$I = (6 \text{ V} - 2 \text{ V}) / 1000 \Omega$$

$$I = 0.004 \text{ A}$$

$$I = 4 \text{ mA}$$

This current level allows the LEDs to illuminate safely without damage.

The timer frequency is controlled by the resistor and capacitor values:

$$f = 1.44 / ((R1 + 2R2) * C)$$

Where:

R represents resistance

C represents capacitance

The capacitors charge and discharge repeatedly, creating the timing signal that drives the LED sequence.

VIII. CONSTRUCTION PROCESS

The construction process began by identifying all components and organizing them according to type and value. Each resistor was measured using a multimeter to confirm its resistance before placement on the board. This helped prevent mistakes and ensured that the correct current-limiting resistors were used for the LEDs.

Before physically assembling the circuit, the design was first created and tested using Multisim software. The LED chaser circuit was built virtually using a 555 timer in astable mode and a 4017 decade counter. The 555 timer was configured to generate clock pulses, and its output was connected to the clock input of the 4017 counter.

Each output pin of the 4017 was then connected to an LED through a 1 kΩ resistor. The simulation allowed me to observe the sequential lighting of the LEDs, which confirmed that the design would work in real life. It also helped me understand how the timing components (resistors and capacitor connected to the 555 timer) controlled the speed of the LED chasing effect.

By adjusting the potentiometer value in Multisim, I was able to see how the flashing speed changed, which made it easier to choose suitable component values before building the actual circuit.

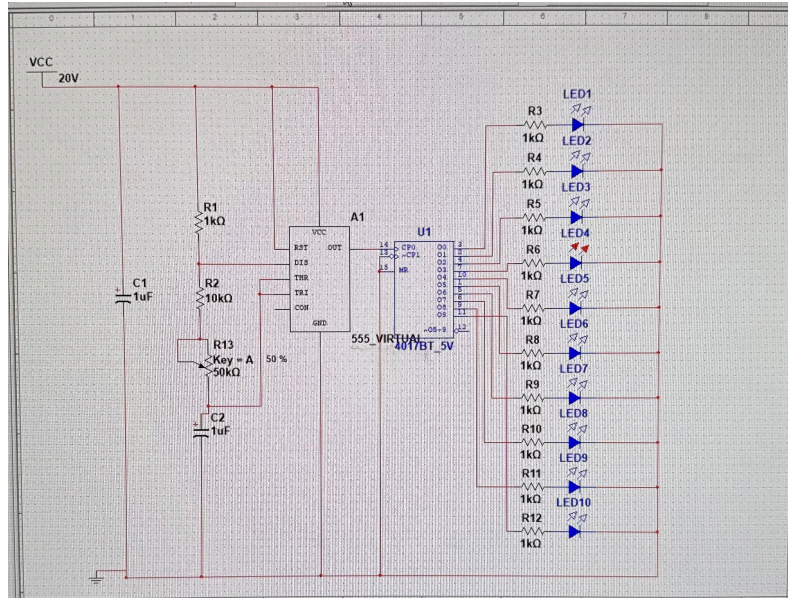


Figure 4: LED Chaser Multisim Diagram Simulation

After confirming the circuit operation in simulation, the physical construction began. The resistors were installed first, followed by the integrated circuits, capacitors, and LEDs. Care was taken to align the integrated circuits with the correct orientation, ensuring that pin 1 matched the marking on the PCB layout.

The LEDs and capacitors required special attention because they are polarized components. The positive (anode) and negative (cathode) leads of the LEDs had to match the PCB markings, and the electrolytic capacitors were inserted with the correct polarity to avoid damage.

After all components were inserted, the leads were bent slightly to hold them in place before soldering.

Soldering was completed on the underside of the board. Each joint was heated properly, and solder was applied carefully to form a strong and reliable electrical connection. After soldering, the board was visually inspected for cold joints or solder bridges. Finally, the power supply was connected and the circuit was tested. The LEDs lit up in sequence as expected, confirming that the construction process was successful.

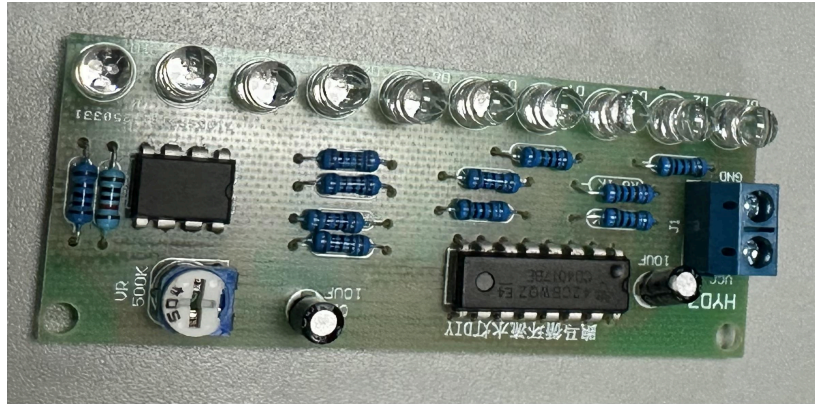


Figure 5. Completed Circuit Front View

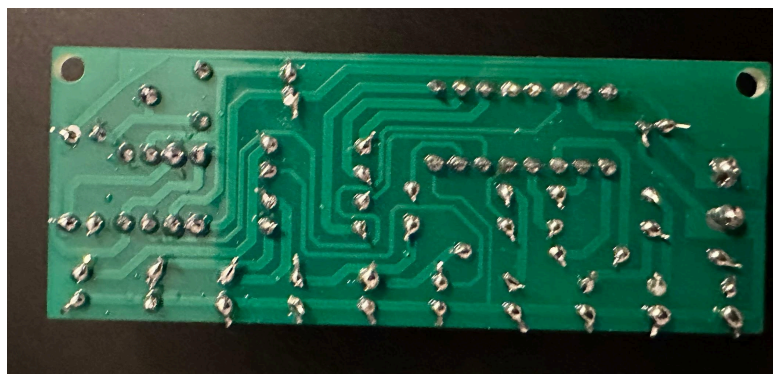


Figure 6. Completed Circuit Back View

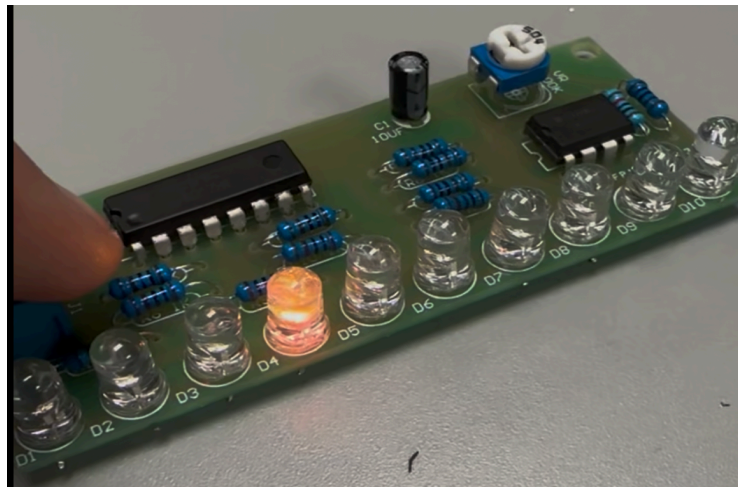


Figure 7. LED Chaser Kit Successful Testing

IX. TROUBLESHOOTING AND TESTING

The circuit did not function correctly on the first or second attempt. Several issues were identified during testing.

The first issue was incorrect polarity of several LEDs, capacitors, Timer IC, and Counter IC. When polarity is reversed, current cannot flow correctly through the component. This prevents the LEDs from lighting and prevents capacitors from charging properly. In some cases, reversed capacitors can become damaged or fail completely.

The second issue was solder bridges forming between connection points. Excess solder created unintended electrical paths that allowed current to flow where it was not supposed to. This prevented the counter integrated circuit from functioning properly.

Another issue was loose components on the printed circuit board. Components that were not secured properly created intermittent connections which caused inconsistent operation.

The board required three separate assembly and testing attempts before the circuit functioned correctly.

Trial 1:

- Incorrect LED polarity
- Incorrect capacitor polarity
- Loose components

Trial 2:

- Solder bridges
- Unintended current paths

Trial 3:

- Correct polarity
- Clean solder joints
- Successful operation

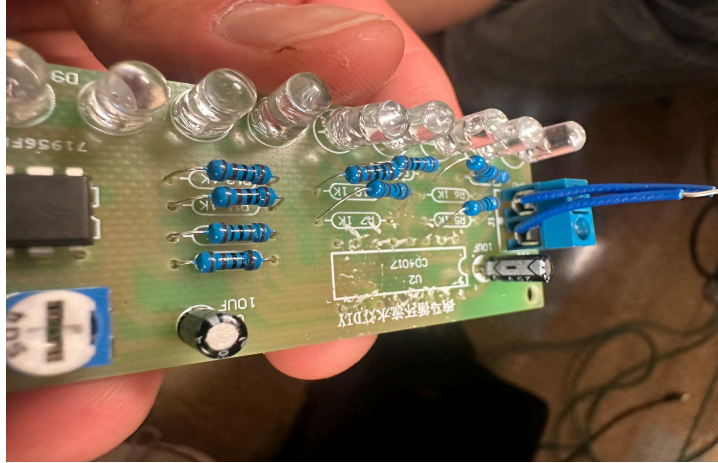


Figure 8. Incorrect Circuit with Attempt to Switch Incorrectly Placed Components

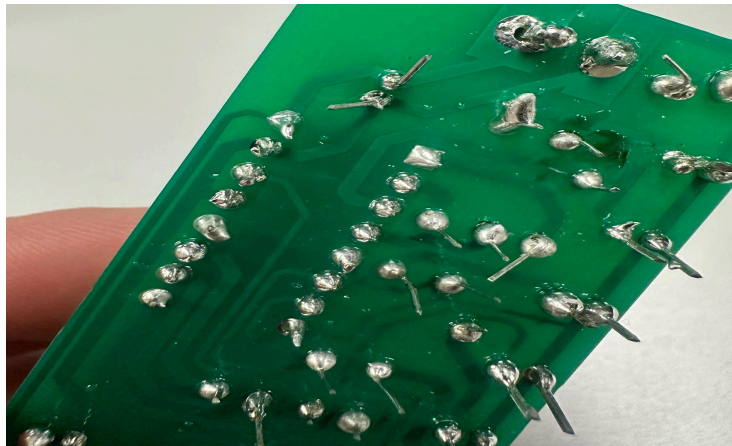


Figure 9. Incorrect Circuit Back View with Overlapping Soldered Components

X. RESULTS

After correcting all errors, the LED chaser circuit functioned as expected. The LEDs illuminated in sequence and repeated continuously.

The timer circuit produced consistent pulses and the counter advanced correctly through each output. All ten LEDs operated correctly and maintained consistent brightness.

XI. DISCUSSION

This project demonstrated the importance of careful assembly and proper soldering techniques. Small mistakes such as incorrect polarity through misplaced components in the wrong direction or excess solder can prevent a circuit from functioning.

The project also demonstrated how multiple components interact within a system. The timer circuit, counter circuit, resistors, and capacitors all play essential roles in producing the final output.

Testing and troubleshooting were important parts of the process. The circuit required multiple attempts before functioning correctly, which reinforced the importance of systematic problem solving.

Understanding current flow and component polarity was critical to completing the project successfully.

XII. CONCLUSION

The LED chaser circuit was successfully constructed and tested. The circuit demonstrated the use of timers, counters, resistors, capacitors, and LEDs in a practical application.

The project provided experience in soldering, circuit construction, and troubleshooting. The final circuit functioned correctly after correcting polarity errors and soldering issues.

This project demonstrated how electronic components work together to create a functional system and provided valuable hands-on experience with electrical circuits.

XIII. FIGURES

Figure 1 – IPO Diagram: LED Chaser

Figure 2 – Network & State Diagram: LED Chaser

Figure 3 – Flow Chart: LED Chaser Operation

Figure 4 – LED Chaser Multisim Diagram Simulation

Figure 5 – Completed Circuit Front View

Figure 6 – Completed Circuit Back View

Figure 7 – LED Chaser Kit Successful Testing

Figure 8 – Incorrect Circuit with Attempt to Switch Incorrectly Placed Components

Figure 9 – Incorrect Circuit Back View with Overlapping Soldered Components

XIV. REFERENCES

1. University of North Texas. (2026). *EENG 1910: Engineering Laboratory course materials*. University of North Texas Canvas.